```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix,precision_score, recall_score,f1_score,roc_curve, roc_auc_score
```

```
data = load_breast_cancer()
```

```
X = pd.DataFrame(data.data, columns=data.feature_names)
y = pd.Series(data.target)
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42
)
```

```
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
```

```
LogisticRegression
LogisticRegression(max_iter=1000)
```

```
y_pred = model.predict(X_test)
y_pred
```

```
array([1, 0, 0, 1, 1, 0, 0, 0, 1, 1, 1, 0, 1, 0, 1, 0, 1, 1, 1, 0, 1, 1,
       0, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1, 0, 1, 1, 0, 1, 1,
       1, 1, 1, 1, 1, 1, 0, 0, 1, 1, 1, 1, 1, 0, 0, 1, 1, 0, 0, 1, 1, 1,
       0, 0, 1, 1, 0, 0, 1, 0, 1, 1, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0, 0, 0,
       1, 1, 1, 1, 1, 1, 1, 1, 0, 0, 1, 0, 0, 1, 0, 0, 1, 1, 1, 0, 1, 1,
       0, 1, 0, 0, 0, 0, 1, 1, 1, 0, 1, 1, 1, 0, 1, 0, 0, 1, 1, 0, 0, 0,
       1, 1, 1, 0, 1, 1, 1, 0, 1, 0, 1])
```

```
accuracy = accuracy_score(y_test, y_pred)
f"accuracy: {round(accuracy,2)}"
```

```
'accuracy: 0.98'
```

```
cm = confusion_matrix(y_test, y_pred)
cm
```

```
array([[53,  1],
       [ 2, 87]])
```

```
precision = precision_score(y_test, y_pred)
f"Precision: {round(precision,2)}"
```

```
'Precision: 0.99'
```

```
recall = recall_score(y_test, y_pred)
f"Recall: {round(recall,2)}"
```

```
'Recall: 0.98'
```

```
f1 = f1_score(y_test, y_pred)
f"f1 score: {round(f1,2)}"
```

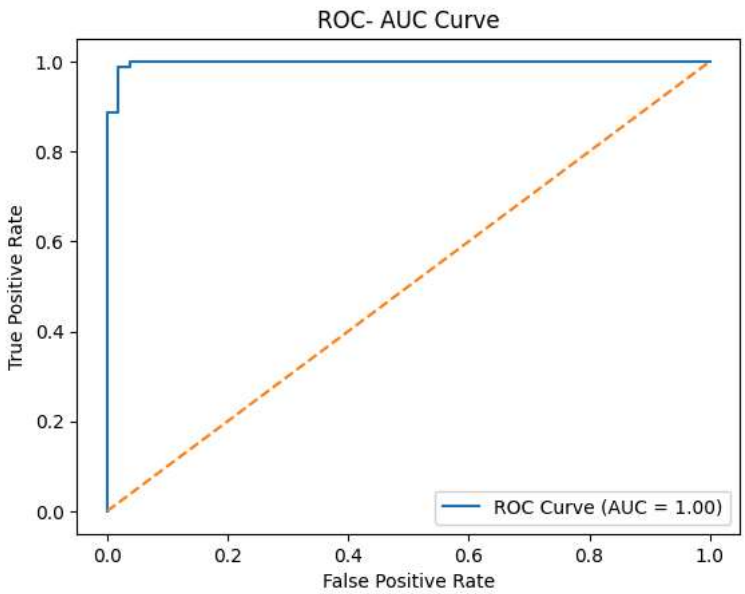
```
'f1 score: 0.98'
```

```
y_prob = model.predict_proba(X_test)[:, 1]

fpr, tpr, thresholds = roc_curve(y_test, y_prob)

roc_auc = roc_auc_score(y_test, y_prob)

plt.figure()
plt.plot(fpr, tpr, label=f"ROC Curve (AUC = {roc_auc:.2f})")
plt.plot([0, 1], [0, 1], linestyle="--")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC- AUC Curve")
plt.legend()
plt.show()
```



Start coding or generate with AI.